Numerical Differentiation.

1) Forward Difference

$$F'(x) = \frac{F(x+h) - F(x)}{h}$$

x	1	1.1	1.2	1.3
$F(x)$	0.000	0.0953	0.1823	0.2624

In here

$$h = x_1 - x_0$$
$$= 1.1 - 1$$
$$= 0.1$$

$$F'(1.1) = \frac{F(1.1+0.1) - F(1.1)}{0.1}$$

$$= \frac{F(1.2) - F(1.1)}{0.1}$$

$$= \frac{0.1823 - 0.0953}{0.1}$$

$$= 0.87$$

original function $F(x) = \ln(x)$

$$F'(x) = \frac{1}{x}$$

$$F'(1.1) = \frac{1}{1.1}$$

$$= 0.91$$

Error $= |f'(x)_{ori} - f'(x)|$

$= 0.91 - 0.87$

$= 0.04$.

2) Backward difference

$$f'(x) = \frac{f(x) - f(x-h)}{h}$$

$$f'(1.1) = \frac{f(1.1) - f(1.1-0.1)}{0.1}$$

$$= \frac{0.0953 - 0.000}{0.1}$$

$$= 0.95$$

Error $= |0.95 - 0.91|$

$= 0.04$.

Central Difference

$$F'(x) = \frac{F(x+h) - F(x-h)}{2h}$$

$$F'(1.1) = \frac{F(1.1+0.1) - F(1.1-0.1)}{2 \times 0.1}$$

$$= \frac{F(1.2) - F(1)}{0.2}$$

$$= \frac{0.1823 - 0.000}{0.2}$$

$$= 0.9115$$

True value = 0.9091

Error = |True value − Approximate value|

= |0.9115 − 0.9091|

= 0.0024.